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How to recover Failed Systemd services automatically

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If you are a Linux user or a System Administrator you must have faced this. This happens many times when we see a service that crashes occasionally on our Linux servers.

In such situations what I don't want is manually start the failed/crashed service and what I want is to make it recover automatically.

In this article, I am going to demonstrate how we can make a Systemd service recover automatically in case of a crash/failure. Before I jump on to the auto-recovery

part for a service let's first recap what you know about Systemd ecosystem.

What is Systemd?

Systemd is a system and service manager for Linux, compatible with SysV and LSB init scripts. **Systemd** creates and manages the sockets used for communication between system components.

What is Systemd Service file?

A unit configuration file whose name ends in ".service" encodes information about a process controlled and supervised by systemd.

What is Systemctl ?

The `systemctl` command is the primary tool to manage *systemd*. It combines the functionality of SysVinit's `service` and `chkconfig` commands into a single tool which you can use to enable and disable services permanently or only for the current session.

Example Systemd Service file

Let's take an example of Systemd service file for sssd daemon.

```
root@ubuntu-server:~# cat /lib/systemd/system/sss.service
[Unit]
Description=System Security Services Daemon
# SSSD must be running before we permit user sessions
Before=systemd-user-sessions.service nss-user-lookup.target
autofs.service
Wants=nss-user-lookup.target

[Service]
ExecStart=/usr/sbin/sss -i -f
Type=notify
NotifyAccess=main
PIDFile=/var/run/sss.pid

[Install]
WantedBy=multi-user.target
```

A Failed Systemd Service

There might be any number of reasons responsible for a service crash or failure such

as misconfigurations, lack of system resources, file/data corruptions etc ..

Here is an example out of a sssd service failure.

```
root@ubuntu-server:~# systemctl status sssd.service
    sssd.service - System Security Services Daemon
    Loaded: loaded (/lib/systemd/system/sss.service; enabled; vendor
preset: ena
    Active: failed (Result: core-dump) since Mon 2020-10-19 15:16:53
UTC; 18min a
    Process: 34488 ExecStart=/usr/sbin/sss -i -f (code=dumped,
signal=ABRT)
    Main PID: 34488 (code=dumped, signal=ABRT)

Oct 18 03:53:14 ubuntu-server sssd[be[34490]: Starting up
Oct 18 03:53:14 ubuntu-server sssd[be[34490]: WARNING: Using a
deprecated filter optio
Oct 18 03:53:14 ubuntu-server sssd[be[34490]: WARNING: Using a
deprecated filter optio
Oct 18 03:53:14 ubuntu-server sssd[34491]: Starting up
Oct 18 03:53:14 ubuntu-server sssd[34493]: Starting up
Oct 18 03:53:14 ubuntu-server sssd[34492]: Starting up
Oct 18 03:53:14 ubuntu-server systemd[1]: Started System Security
Services Daemon.
Oct 19 15:16:53 ubuntu-server systemd[1]: sssd.service: Main process
exited, code=dump
Oct 19 15:16:53 ubuntu-server systemd[1]: sssd.service: Unit entered
failed state.
Oct 19 15:16:53 ubuntu-server systemd[1]: sssd.service: Failed with
result 'core-dump'
```

How to make Systemd service auto recover itself ?

To achieve auto recovery add following two options in your systemd service file.

To edit Systemd service file run following command:

```
root@ubuntu-server:~# systemctl edit sssd.service
```

And add these two parameters under the service section:

Restart=on-failure

RestartSec=30s

In a systemd service unit file, the two parameters `Restart` and `RestartSec` controls crash behavior. The first parameter specifies when the service should restart, and the second parameter defines how long it should wait before restarting.

Restart =

Configures whether the service shall be restarted when the service process exits, is killed, or a timeout is reached.

Possible values: `no`, `on-success`, `on-failure`, `on-abnormal`, `on-watchdog`, `on-abort`, or `always`.

If set to `no` (the default), the service will not be restarted.

If set to `on-success`, it will be restarted only when the service process exits cleanly.

If set to `on-failure`, the service will be restarted when the process exits with a non-zero exit code, is terminated by a signal (including on core dump, but excluding the aforementioned four signals)

If set to `always`, the service will be restarted regardless of whether it exited cleanly or not, got terminated abnormally by a signal, or hit a timeout.

RestartSec =

Configures the time to sleep before restarting a service (as configured with `Restart=`). Takes a unit-less value in seconds, or a time span value such as "5min 20s". Defaults to 100ms.

```
root@ubuntu-server:~# cat /lib/systemd/system/sss.service
[Unit]
Description=System Security Services Daemon
# SSSD must be running before we permit user sessions
Before=systemd-user-sessions.service nss-user-lookup.target
autofs.service
Wants=nss-user-lookup.target

[Service]
ExecStart=/usr/sbin/sss -i -f
Type=notify
NotifyAccess=main
```

```
PIDFile=/var/run/sss.pid
Restart=on-failure
RestartSec=30s

[Install]
WantedBy=multi-user.target
```

NOTE: Exit causes and the effect of the `Restart=` settings on them

Restart settings/Exit causes	no	always	on-success	on-failure	on-abnormal	on-abort	on-watchdog
Clean exit code or signal		X	X				
Unclean exit code		X		X			
Unclean signal		X		X	X	X	
Timeout		X		X	X		
Watchdog		X		X	X		X

Source: <https://www.freedesktop.org/software/systemd/man/systemd.service.html>

Reload the Systemd daemon

If you update Systemd service file manually you must make *systemd* aware of the updated service. to do that you need to reload its service files.

```
root@ubuntu-server:~# systemctl daemon-reload
```

That's all!

Hope you like the article. Please let me know your feedback in the response section.

Thanks. Happy learning!



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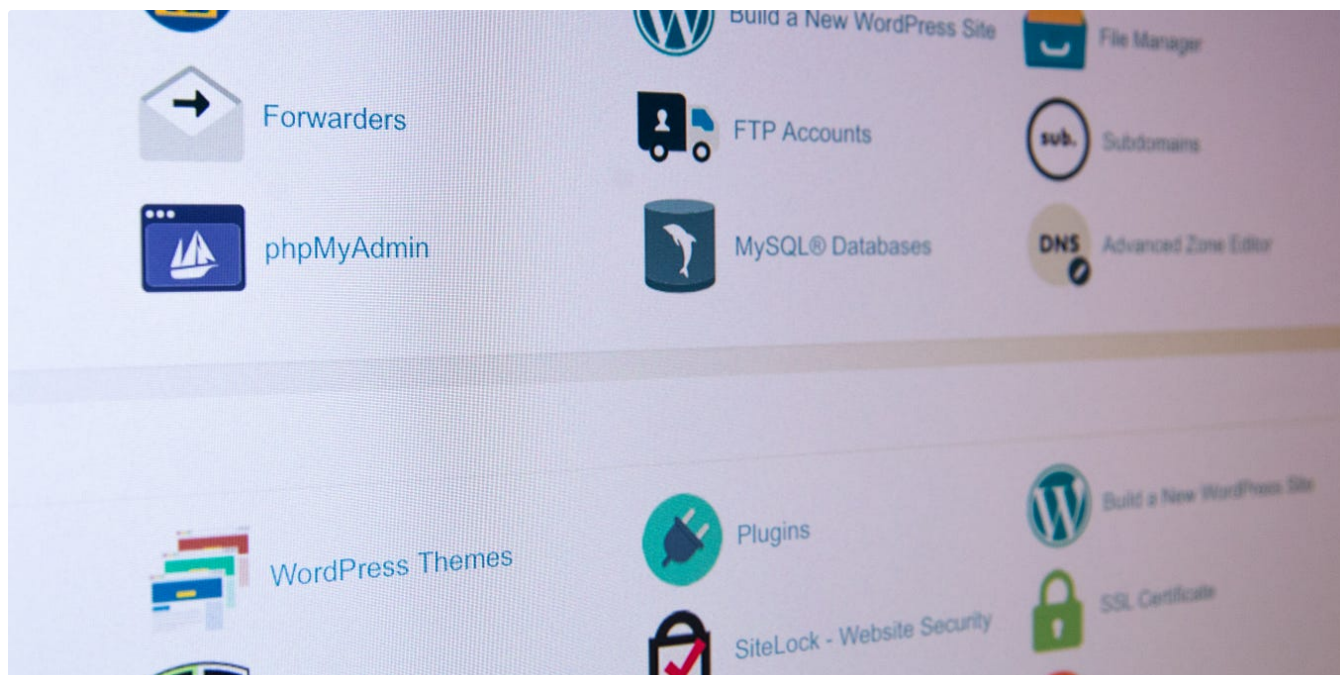


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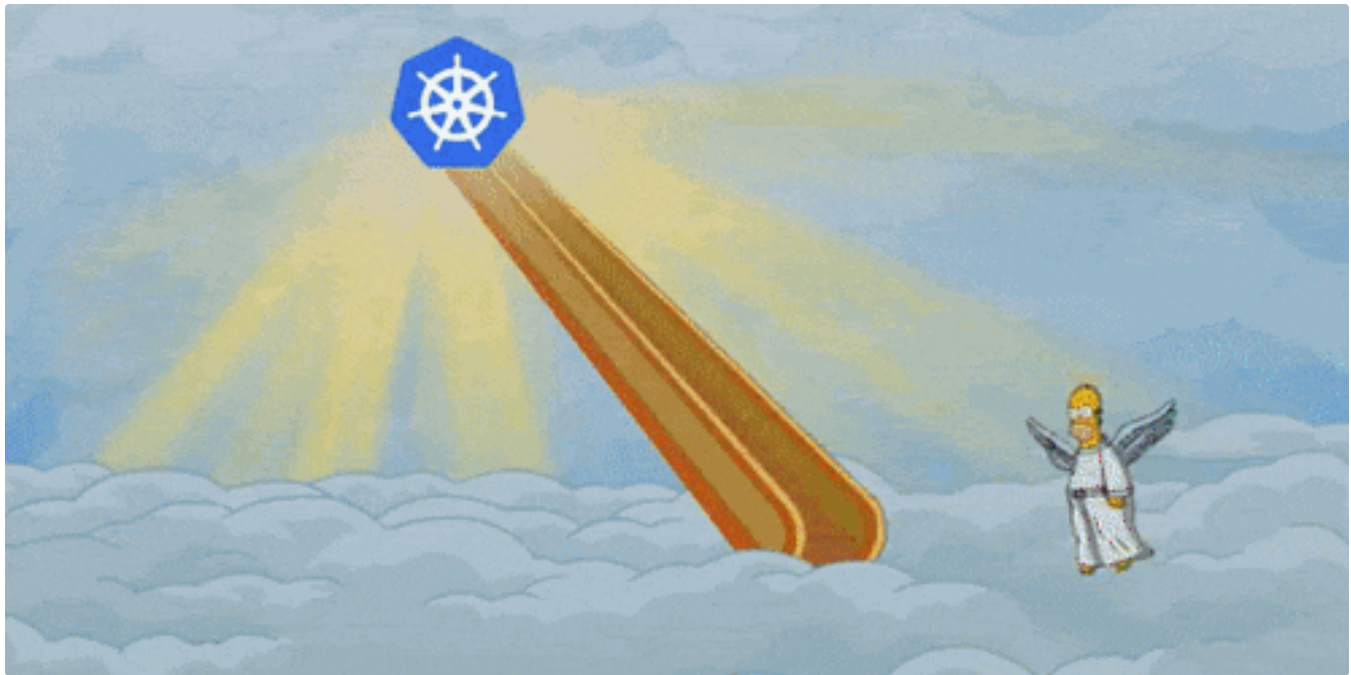
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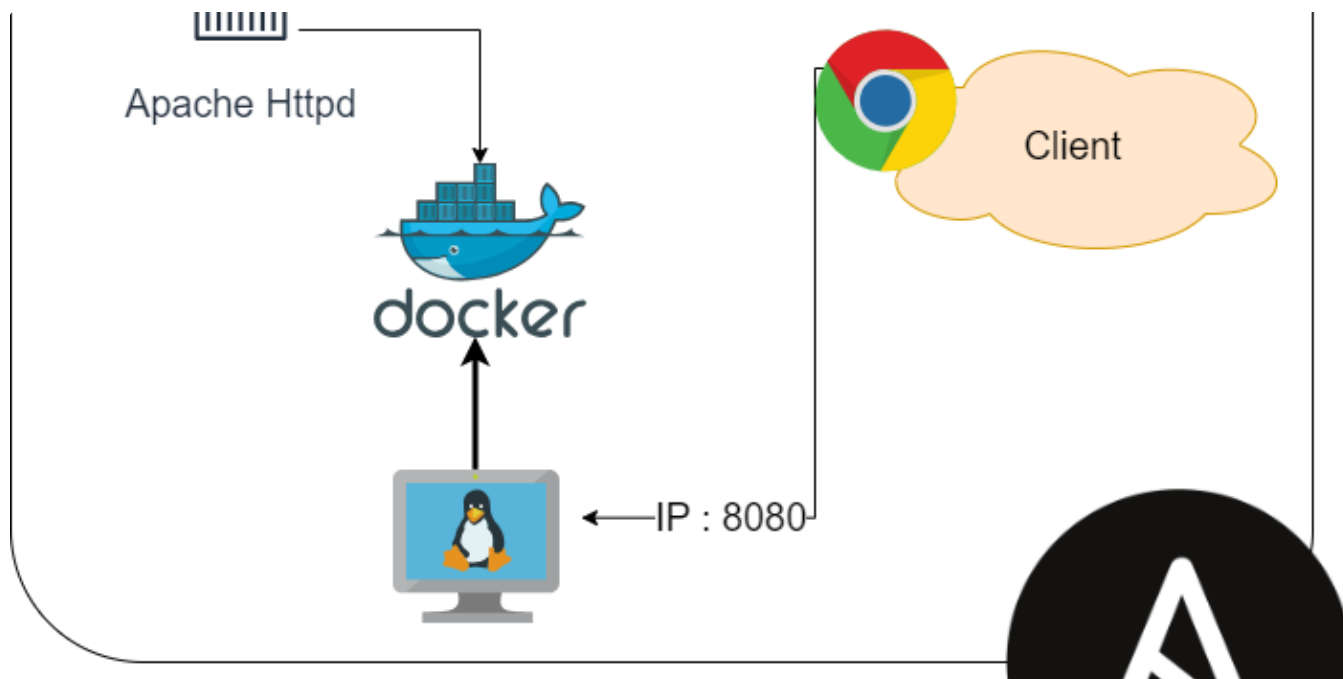
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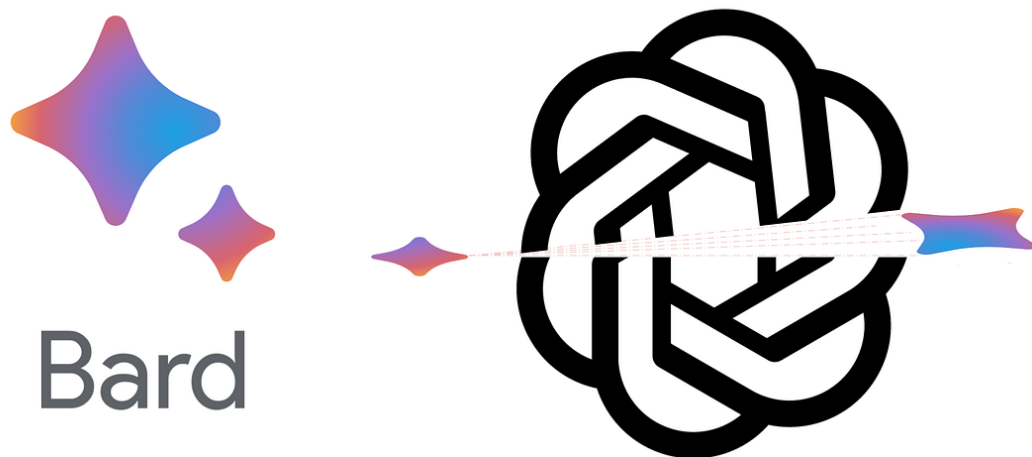
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